Financial Management

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1. Foundation of Financial Management: Time Value of Money

1.5 Time Value of Money

Money has time value. A rupee today is more valuable than a rupee a year hence.

Why??????

There are several reasons:

- Individuals, in general, prefer current consumption to future consumption.
- Capital can be employed productively to generate positive returns.
- In an inflationary period a rupee today represents a greater real purchasing power than a rupee a year hence.

Value of money today is not equal to value of money tomorrow

Inflation

Measure money = quantity it can purchase

Rs. 100 = 3 items at t=0

Rs. 120 = 3 items same quantity at t=2

Investment ➔ multiplied at a certain rate => interest rate

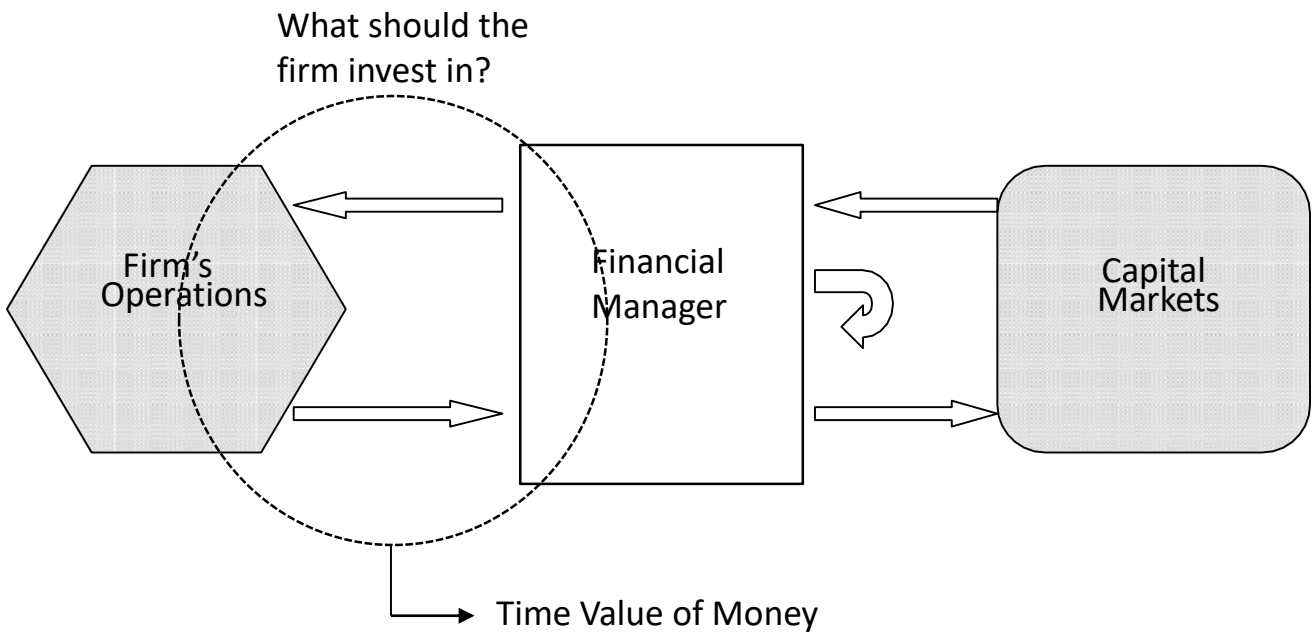
Rs. 100000 @ 7% FD for 5 years ➔100000*(1+0.07)^5 = 140255

Inflation rate 6.5% for next 5 years

Effective rate of interest = Interest rate – Inflation rate = 0.5%

Time Value of Money

- We begin by addressing "***What should the firm invest in?"*** (the investment decision) within the finance function.



Time value of money

- How much it costs to rent money
  - Keeping the money at home creates an opportunity cost
  - Interest rate is a measure of this opportunity cost
  - Price of impatience and the value of waiting

A dollar today is worth more than a dollar tomorrow

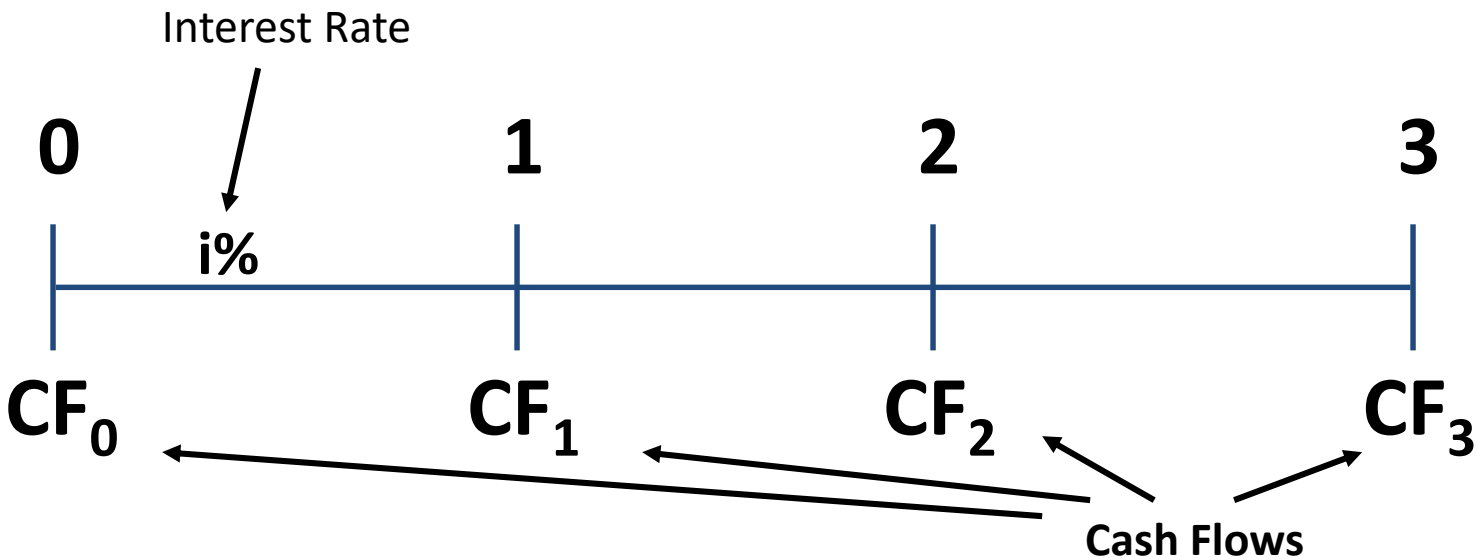
- Example: Assume you have $100 and interest rate is 10%
  - PV = Rs.100

- At the end of year 1, FV = PV + r(PV) = PV(1+r) = Rs.110; Where:
  - FV = Future Value
  - PV = Present Value
  - r = interest rate (or discount rate)
- At the end of year 2, FV = Rs.110 x 1.1 = $121
  - Interest on interest = Rs.1 = Compound Interest

- In general, FV = PV(1+r)^n; where n = number of time periods

- Reversing the logic: PV = FV/ (1+r)^n

Time lines show timing of cash flows.



<u>Tick marks</u> at ends of periods.

- Time 0 is today; Time 1 is the end of Period 1; or the beginning of Period 2.

90% of getting a Time Value problem correct is setting up the timeline correctly!!!

There are 3 ways of looking at Interest rates

- Returns required by investors to induce them to postpone consumption
  - Rate of interest that can convert consumers into investors or lenders
- Discount rate
  - Generic term that measures the rate of return reflecting the time value of money
- Opportunity cost: value of next best option that investors forgo by choosing what they do

Opportunity cost is a basic economics concept

- Opportunity Cost
  - Difference between a particular action and the value of the best alternative
  - Explicit recognition of the reality that taking an action eliminates other avenues/ options
  - Indication of the relative importance of a decision

- If opportunity cost is small => cost of incorrect choice is small
- If opportunity cost is large => cost is large

- Examples
  - buying your groceries (likely small)
  - buying a car (potentially large)

- However, it is not always obvious or easy to calculate

Inflation and risk impact interest rates

- In a certain world interest rate is the risk free rate
  - Closest is government short term debt like US T-bills
- In real world however two factors come into play
  - 1) Inflation: lenders charge an inflation premium to account for expected increase in prices
  - 2) Risk: Compared to governments there is risk inherent in lending to private companies, driven primarily by business dynamics such as industry climate, competition, economy, etc.
    - This is typically referred to as default risk
- Interest rates are also affected by the forces of demand and supply

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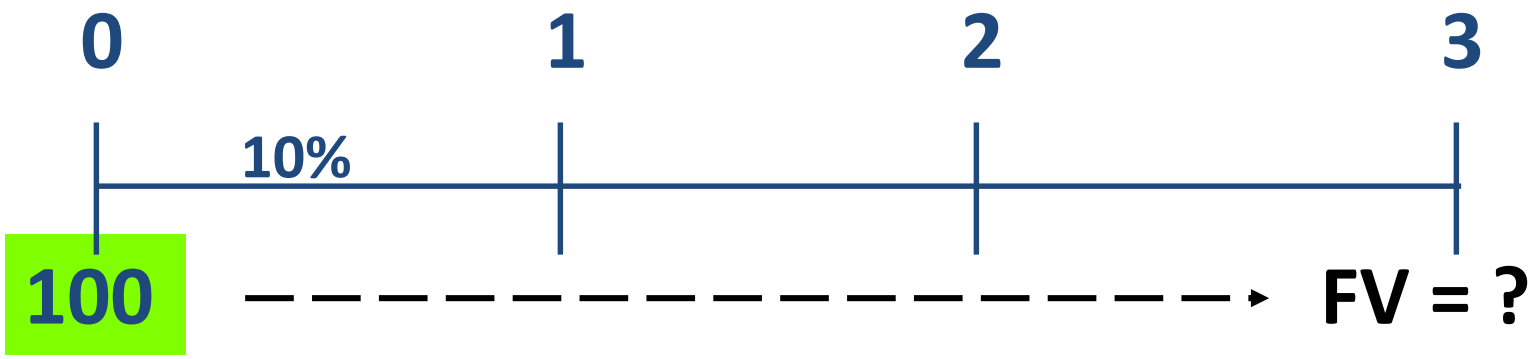
FUTURE VALUE

Future Values

What's the FV of an initial Rs.100 after 3 years if i = 10%?



Finding FVs (moving to the right on a time line) is called compounding.

- **Compounding involves earning interest on interest for investments of more than one period.**

Future Value of a Single Amount

$$\boldsymbol{FV_n} = \mathbf{PV}(\mathbf{1} + \boldsymbol{k})^{\boldsymbol{n}}$$

Where

FV_n = future value *n* years hence
PV = cash today (present value)
k = interest rate per year
n = number of years for which compounding is done

This is the basis equation in compounding analysis. The factor $(1 + k)^n$ is referred as the compounding factor or the future value interest factor $(FVIF_{k,n})$

Value of ($FVIF_{k,n}$)

Example: If you deposit Rs. 1000, today in a bank which pays 10% interest compounded annually, how much will the deposit grow to after 8 years and 12 years?

FV_n = PV$(1+k)^n$

$F1000_8$ = Rs. 1000 $(1+0.10)^8$ = Rs. 2144

$F1000_{12}$ = Rs. 1000 $(1+0.10)^{12}$ = Rs. 3138

Double Period

Rule of 72 : According to this rule of thumb the doubling period is obtained by dividing 72 by the interest rate. Eg. if the interest rate is 8%, the doubling period is about 9 years = 72/8

Rule of 69: According to this rule of thumb, the doubling period is equal to = 0.35 + (69/Interest Rate). Eg. if the interest rate is 10%, the doubling period is 0.35+(69/10) = 7.25 years.

Finding the Growth Rate

To calculate the compound rate of growth of some series eg. sales series or the profit series, over a period of time, $(FVIF_{k,n})$ can be used.

Example: Alpha Ltd. is having following details.

Years	2011	2012	2013	2014	2015	2016	2017
Sales (in millions)	50	57	68	79	86	92	99

Step1: Find the ratio of sale of 2017 to 2011 = 99/50 = 1.98

Step 2: Refer the $(FVIF_{k,n})$ table and look at the row to find out the value closest to 1.98 and then read the interest rate corresponding to this. The growth rate is 12%

Shorter Compounding Period

$$FV_n = PV(1 + k/m)^{mXn}$$

Where

FV_n = future value *n* years hence

PV = cash today (present value)

k = nominal annual rate of interest

n = number of years for which compounding is done

m = number of times compounding is done during a year

Value of Shorter Compounding Period

Example: How much does a deposit of Rs. 5000 grow to at the end of 6 years, if the nominal rate of interest is 12% and the frequency of compounding is 4 times a year?

The amount after 6 years = $\text{Rs. } 5000\ (1 + 0.12/4)^{4X6}$

= Rs. 5000 $(1.03)^{24}$

= Rs. 5000 X 2.0328

= Rs. 10164

Effective Annual Rate

- With all the possible interest rates one can use, we need a practical method for comparing them.
- Numerical Example:
  - Suppose that you invest Rs.100 at an annual rate of 10% compounded semi-annually.
  - After one year, you will have

$$\mathrm{R}s.100(1+\frac{10\%}{2})^2 = Rs.110.25$$

  - **Question:** What is the annual interest rate (compounded annually) that would generate the same amount after one year? Notice that the ***amount of interest doesn't change***, just how we "***quote***" it.

Effective Annual Rate:

Numerical Example (continued):

– We need to solve:

$$\text{Rs.}\,100(1+\frac{\hat{r}}{1})^{1*1} = 110.25 \xrightarrow{\text{yields}} \hat{r} = 10.25\%$$

- The future value of Rs.100 invested at an annual rate of *10% compounded semi-annually* is the same as the future value of Rs.100 invested at an annual rate of *10.25% compounded annually*.

- This 10.25% has many names. It is called the ***effective annual rate***, equivalent annual rate, annual percentage rate or ***annual percentage yield (APY).***

Exercise on Compounding and Effective Rates

- You want to invest Rs.1,000 for two years. After visiting three different banks, you discover that you have three possible options:

  (i) an annual interest rate of 12.5%, compounded annually

  (ii) an annual interest rate of 12%, compounded quarterly

  (iii) a continuously compounded rate of 11.75%.

  – Which savings account should you choose?
  – What is the effective monthly rate that you will then be earning?

Solution

In two years, the three different accounts would respectively accrue to:

(i) $1{,}000(1+0.125)^2 = 1{,}265.63$

(ii) $1{,}000(1+\frac{0.12}{4})^{4*2} = 1{,}266.77$

(iii) $1{,}000e^{0.1175*2} = 1{,}264.91$

You should therefore invest in the quarterly compounded account (ii). Note that you could also have solved this problem by calculating future value factors for all three rates, or by comparing equivalent annual rates as below:

(i) Obviously, $\tilde{r} = 12.50\%$

(ii) $1 + \tilde{r} = (1+\frac{0.12}{4})^4 \xrightarrow{\text{yields}} \tilde{r} = 12.55\%$

(iii) $1 + \tilde{r} = e^{0.1175*1} \xrightarrow{\text{yields}} \tilde{r} = 12.47\%$

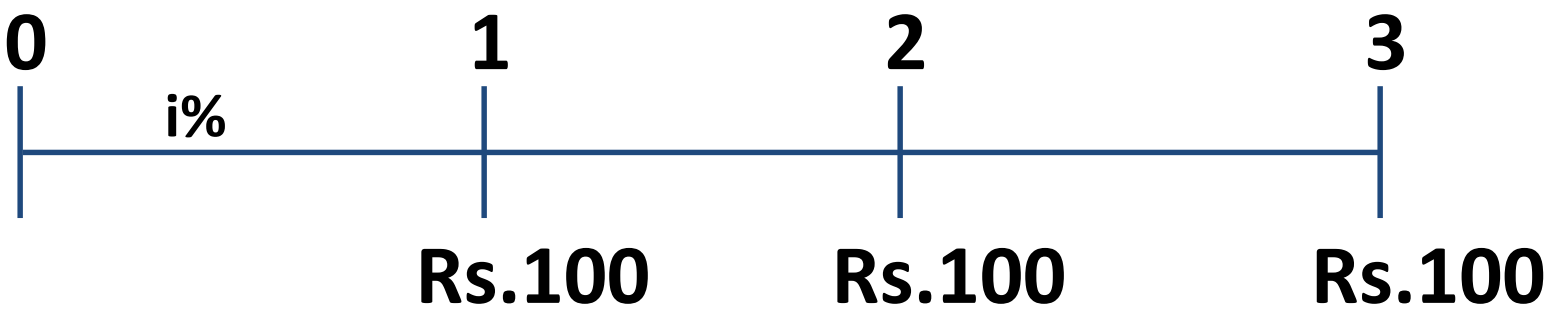
– Clearly, the second bank offers the best deal.

Future Value of An Annuity

- An annuity is a series of periodic cash flows of equal amounts. Eg. the premium payment of life insurance policy.
- **When the cash flows occur at the end of each period the annuity is called a regular or ordinary or deferred annuity.**
- **When the cash flows occur at the beginning of each period the annuity is called an annuity due.**

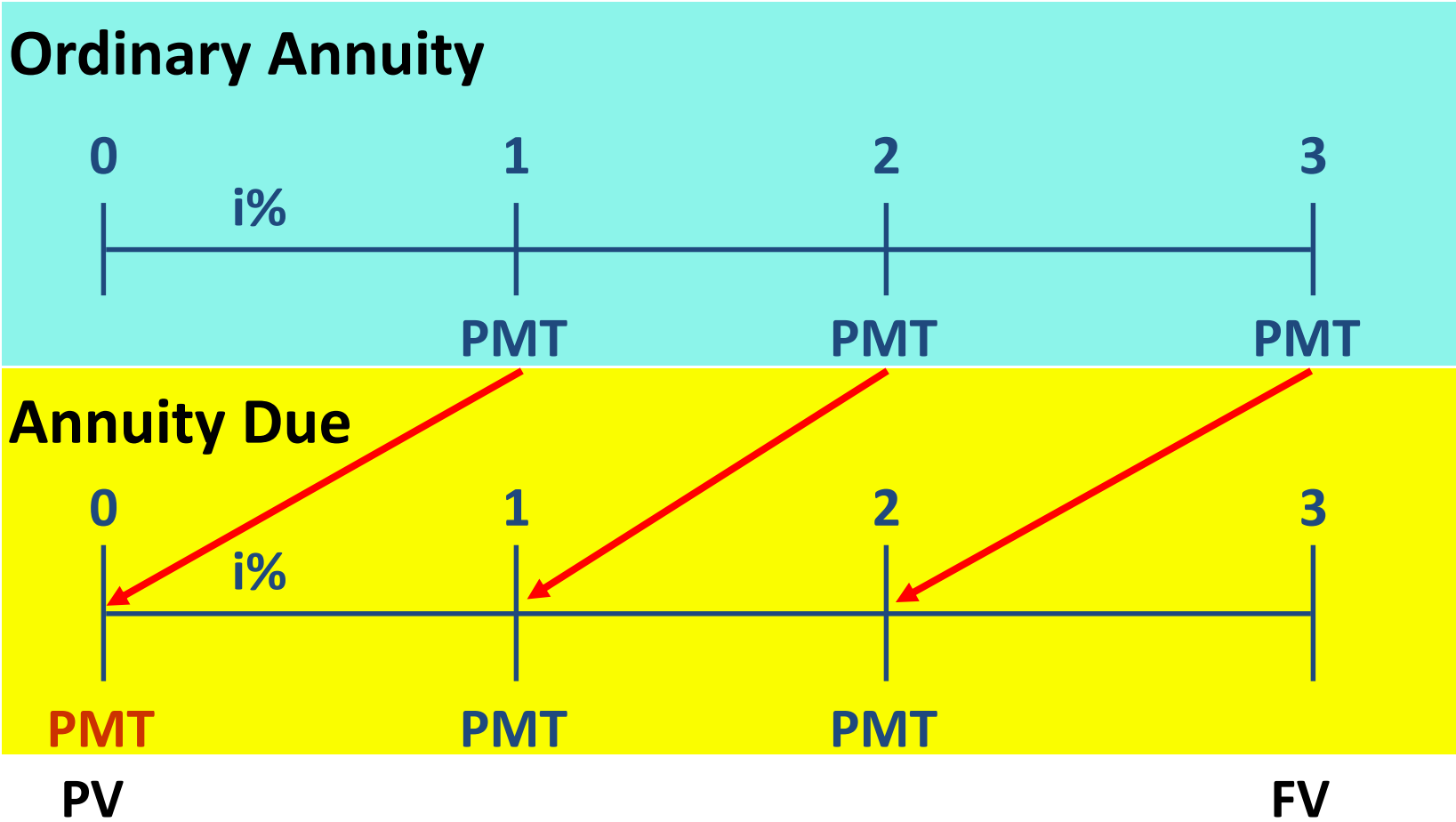
Ordinary Annuity Timeline

Time line for an ordinary annuity of Rs.100 for 3 years.



Ordinary Annuity vs. Annuity Due

Difference between an <u>ordinary</u> <u>annuity</u> and an <u>annuity due</u>?



Future Value of An Ordinary Annuity - Formula

$$FVA_n = \text{PMT X} \left[\frac{(1+r)^n - 1}{r}\right]$$

Where

FVA_n = future value of an annuity which has a duration of *n* period

PMT = payment amount per period

r = interest rate per period

n = duration of the annuity

$\left[\frac{(1+r)^n - 1}{r}\right]$ is referred as the future value interest factor for an annuity.

Future Value of An Annuity

Example 1: Four equal annual payments of Rs. 2000 are made into a deposit account that pays 8% interest per year. What is the future value of this annuity at the end of 4 years.

$$\boldsymbol{FVA_n} = \mathrm{Rs.}\,2000\left[\frac{(1+0.08)^4-1}{0.08}\right] = \text{Rs. } 2000(4.506) = \text{Rs.}9012$$

Example 2: You want to buy a house after 5 years when it is expected to cost Rs. 2 million. How much should you save annually if your savings earn a compound return of 12%?

$$\text{Cf} = \mathrm{Rs.}\,2000000 \,/\left[\frac{(1+0.12)^5-1}{0.12}\right] = \text{Rs.}2000000/(6.353)$$

$$= \text{Rs.}314812$$

Future Value of An Annuity Due- Formula

$$FVA_n = \text{PMT X} \left[\frac{(1+r)^n - 1}{r}\right] \text{X (1+ r)}$$

Where

FVA_n = future value of an annuity which has a duration of *n* period

PMT = payment amount per period

r = interest rate per period

n = duration of the annuity

$\left[\frac{(1+r)^n - 1}{r}\right]$ is referred as the future value interest factor for an annuity.

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PRESENT VALUE

Present Value of a Single Amount

$$\text{PV} = FV_n X \left[\frac{1}{1+r}\right]^n$$

$\left[\frac{1}{1+r}\right]^n$ **is called Present Value Interest Factor**

Example: find the present value of Rs. 1000 receivable 6 years hence if the rate of discount is 10%.

PV = $1000_6 \left[\frac{1}{1+0.10}\right]^6$ = Rs. 1000 X 0.564 = Rs. 564

Present Value of an Uneven Series

$$\mathbf{PV}_n = \sum_{t=1}^{n} \left[\frac{PMT_t}{(1+r)^t} \right]$$

Where

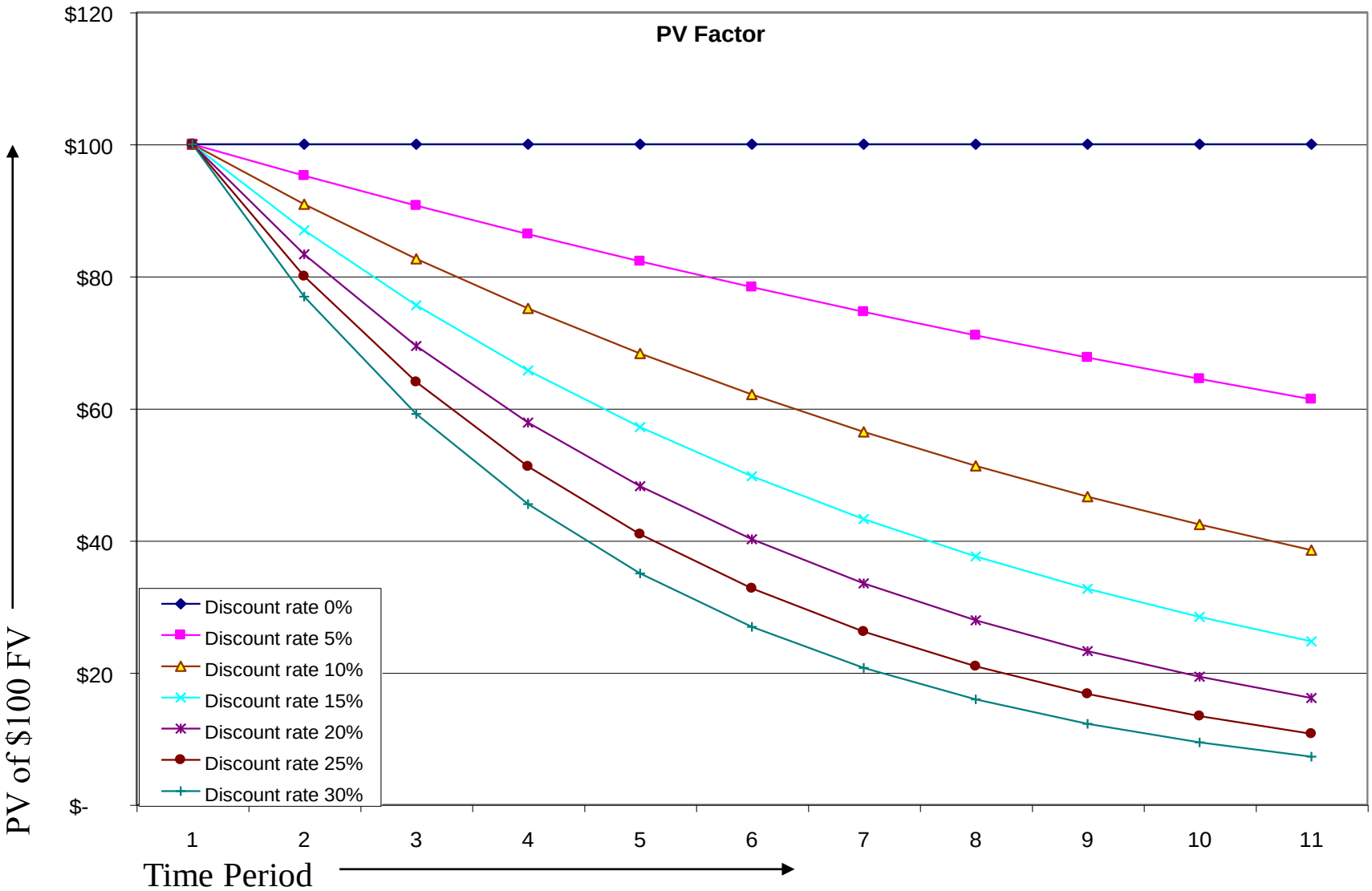
PV_n = Present value of a cash flow stream

PMT_t = Cash flow occurring at the end of year t also know as CFt

r = Discount rate also denoted as k

n = Duration of the cash flow stream

In calculating PV, choice of discount rate is fairly critical



Present Value of an Uneven Series

Example: Calculate present value of the following cash flow stream if the discount rate is 12%

Year	1	2	3	4	5	6	7	8
Cash Flow	1000	2000	2000	3000	3000	4000	4000	5000

Shorter Discounting Period

$$\mathbf{PV} = \mathbf{F}V_n \left[\frac{1}{1+k/m}\right]^{mXn}$$

Where

PV = present value

FV_n = cash flow after *n* years

m = number of times per year discounting is done

n = duration of the cash flow stream

k = annual discount rate

Shorter Discounting Period

Example: A cash flow of Rs. 10000 to be received at the end of four ears. Determine the present value of this cash flow when the discount rate is 12% and discounting is done quarterly.

$$\mathbf{PV} = \mathbf{F}\boldsymbol{V_n}\left[\frac{\mathbf{1}}{\mathbf{1}+\boldsymbol{k}/\boldsymbol{m}}\right]^{\boldsymbol{mn}} = \text{Rs. } 10000 \text{ X } PVIF_{\frac{k}{m},mXn}$$

$$= \text{Rs. } 10000 \text{ X } PVIF_{\frac{0.12}{4},4X4}$$

$$= \text{Rs. } 10000 \text{ X } PVIF_{0.3,16}$$

$$= \text{Rs. } 6230$$

Car Purchase Problem:

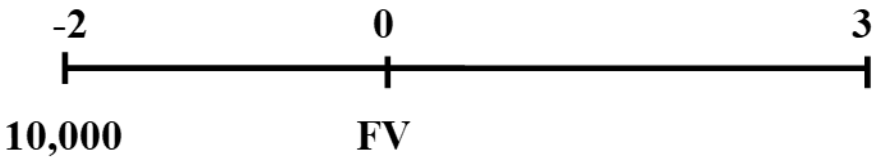
- Two years ago, you put $10,000 dollars in a savings account earning an annual interest rate of 8% compounded semiannually. At the time, you thought that these savings would grow enough for you to buy a new car five years later (i.e. three years from now).
- However, you just re-estimated the price that you will have to pay for the new car in three years at $18,000.

(i) How much more money do you need to put in your savings account now for it to grow to this new estimate in three years?

(ii) Now suppose the car company will offer you to pay for the car over some time. In particular, you will have the opportunity to make a down payment of $6,000 at the time you get the car (three years from now) and to make additional payments of $6,500 at the end of each of the following two years. With this new offer, how much money do you need to add to your account now?

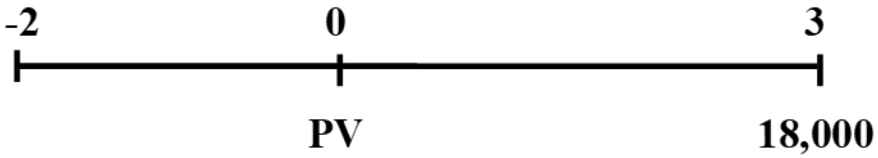
Solution

- Let us first figure out how much money FV is in the savings account now.



$$FV = \$10{,}000 * \left(1 + \frac{0.08}{2}\right)^{2*2} = \$10{,}000 * (1 + 0.04)^4 = \$11{,}698.59$$

- Now, the account should have an amount PV in it for it to grow to $18,000 in three years.



$$PV = \frac{\$18{,}000}{\left(1 + \frac{0.08}{2}\right)^{2*3}} = \frac{\$18{,}000}{(1.04)^6} = \$14{,}225.66$$

- Therefore, you need to put the difference, ($14,225.66 - $11,698.59) = $2,527.07 in the account today.

Solution

- Considering the financing option, the present value (at time 0) of the three payments is:

$$PV = \frac{\$6{,}000}{\left(1+\frac{0.08}{2}\right)^{2*3}} + \frac{\$6{,}500}{\left(1+\frac{0.08}{2}\right)^{2*4}} + \frac{\$6{,}500}{\left(1+\frac{0.08}{2}\right)^{2*5}} = \$13{,}882.54$$

- Thus, you need to add $13,882.54 - $11,698.59 = $2,183.95 to the account.

- ***Question:*** Can you see a problem or difficulty in doing calculations this way?

Present Value of Ordinary Annuity

$$PVA_n = \text{PMT X} \left[\frac{1-\frac{1}{(1+r)^n}}{r}\right]$$

PVA_n = present value of an annuity which has a duration of *n* period

PMT = constant periodic cash flow

r = discount rate

$\left[\frac{1-\frac{1}{(1+r)^n}}{r}\right]$ is referred to as the present value interest factor for an annuity ($PVIFA_{r,n}$)

Present Value of Annuity Due

$$PVA_n = PMT \times \left[\frac{1-\frac{1}{(1+r)^n}}{r}\right] \times (1+r)$$

PVA_n = present value of an annuity which has a duration of *n* period

PMT = constant periodic cash flow

r = discount rate

$\left[\frac{1-\frac{1}{(1+r)^n}}{r}\right]$ is referred to as the present value interest factor for an annuity ($PVIFA_{r,n}$)

Present Value of Annuity

Example: what is the present value of a 4 year annuity of Rs. 10000 discounted at 10%?

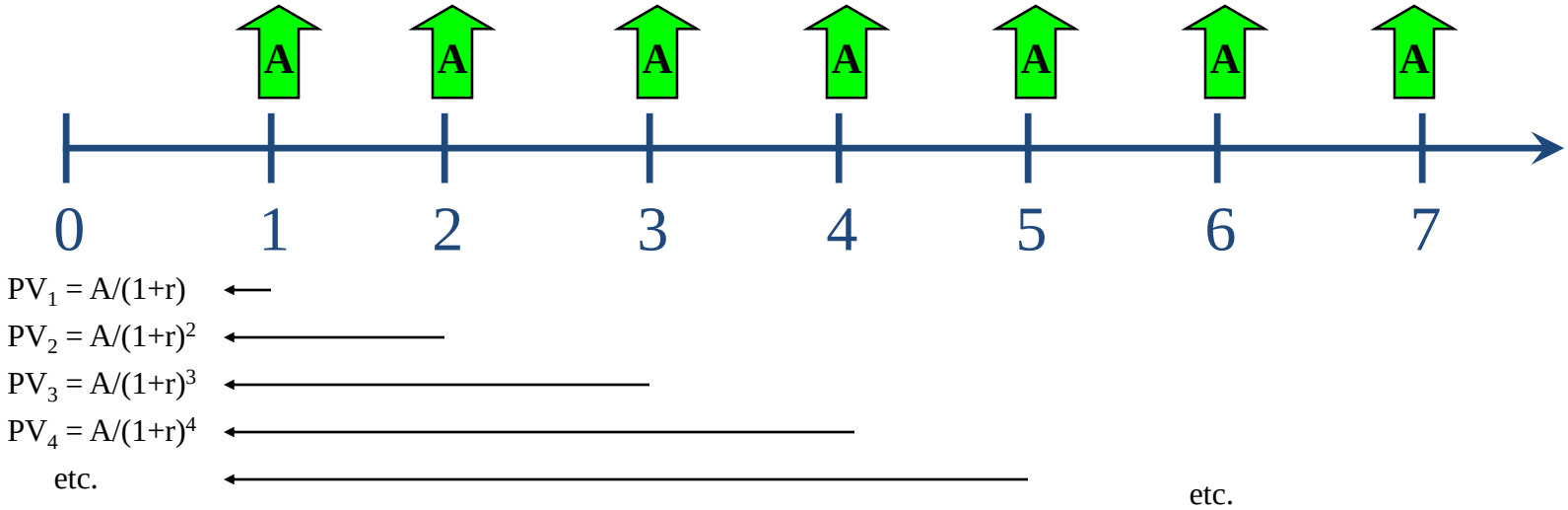
P$\boldsymbol{VA_n}$ **= PMT X (**$\boldsymbol{PVIFA_{k,n}}$**)**

= Rs. 10000 X 3.170

= Rs. 31700 (ordinary annuity)

Perpetuities

Perpetuity is a series of constant payments, A, each period forever.



$$PV_{perpetuity} = \sum[A/(1+i)^t] = A \sum[1/(1+i)^t] = A/i \text{ or } CF/r$$

<u>Intuition:</u>

Present Value of a perpetuity is the amount that must invested today at the interest rate *i* to yield a payment of *A* each year without affecting the value of the initial investment.

Perpetuity Example: UK Consols

- In the 1800's, the British government decided to consolidate the huge debt accumulated during the Napoleonic wars and to replace it with a single issue of bonds with no termination date and a coupon rate of 2½%. These bonds, called ***consols*** and they still exist today.

- Suppose that the current interest rate in the U.K. is 9%. What is the present value of a consol with a £ 1,000 face value?
  - This is just a perpetuity promising to pay £ 25 each year. Its present value is:

$$PV = \frac{£\,25}{0.09} = £\,277.78$$

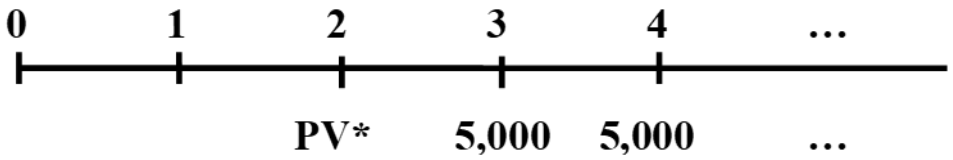
******Caution:*** Don't confuse a coupon rate and the current market interest rate, they are different!

Deferred Perpetuity Problem

- A rich entrepreneur would like to set up a foundation that, every year, will pay $5,000 in the form of a scholarship to one deserving University Business student.
  - The first such scholarship is to be awarded *in three years*, and a scholarship will be awarded in *perpetuity* every year after that.
  - How much money should the entrepreneur put in the foundation's account, if that account earns 8% compounded annually?

Solution

- First, let us calculate how much money will need to be in the account ***at the end of the second year***; let us denote that amount by PV*.



- For the account to be worth this much in two years, the amount that the entrepreneur needs to contribute ***initially*** is:

$$PV = PV^* \frac{1}{(1+0.08)^2} = \frac{5{,}000}{0.08}\left(\frac{1}{(1.08)^2}\right) = \$53{,}583.68$$

- Notice that the perpetuity starts in ***three*** years, but the exponent on the discount factor is a ***two***.

- This is because the perpetuity formula used in the first step calculates the value of a stream of cash flows starting a year later. Thus, ***only two more years*** of discounting are needed.

Growing Perpetuity

– ***Definition:*** A ***growing perpetuity*** is a set of constant cash flows CF that grows at a rate of g every year forever, where the amount in year t is $C(1+g)^{t-1}$

$$PV = \frac{CF}{r-g}$$

Growing Perpetuity Example: Endowed Chair

- A benefactor wishes to endow a Chair in Finance at the BITS Pilani. The aim is to provide an amount equaling $150,000 in the first year and growing at a rate of 5% each subsequent year in order to adjust for the expected growth in salaries. Suppose that the interest rate is 10%. How much should the benefactor donate?
  - We have r = 10%, g = 5%, C = 150,000. Therefore,

$$PV = \frac{C}{r-g} = \frac{\$150{,}000}{0.10 - 0.05} = \$3{,}000{,}000$$

Equated Monthly Instalment

EMI can be calculated using following formula:

$$PVA_n = [P \text{ X } r \text{ X } (1+r)^n]/[(1+r)^n - 1]$$

Where

P = principal amount

r = monthly interest rate

n = loan tenure in months

EMI Example

Shyam takes a housing loan of Rs. 1000000 carrying an interest of 1 per cent per month. The loan is to be repaid over 180 months. What is the EMI?

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Valuation: Applications of TVM in Security (Bonds & Equity Share) Analysis

Equity Valuation

1. Face value = issue price at the time of IPO
2. Market value of equity = extrinsic value of the equity = observed value = No. of equity share * CMP
3. Calculated value of equity = intrinsic value of equity. = Time value of money

Projection of FCF = DCF

Estimated cost of the project = Rs. 500 crores = PV = Cash outflow

Estimated production, sales and revenue

Estimated expenses including depreciation

Estimated net profit

Estimated cash inflow = Free Cash flow (FCF) = NP + Depreciation + change in investments (change in FA & change in WC)

FCF will be generated over a life of the project. Forecasting will help in estimating life of the project.

Decide the discount factor – based on current market interest rate

Discount all the FCF with this discount factor to get PV of Future FCF

Sum of all the PV of FCF and compare it with PV = Cash outflow

Sum of PV of FCFs – Cash outflow

If it is positive = invest = accept

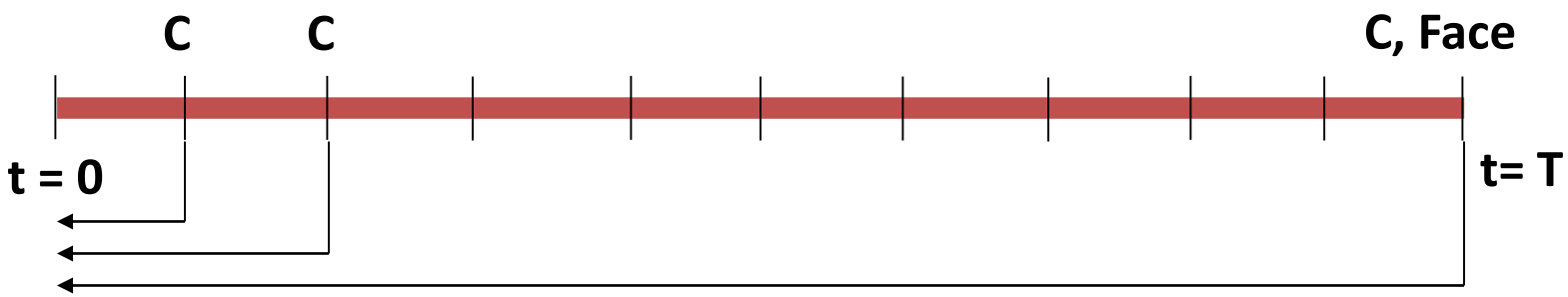
If negative = reject

If zero = may go for it.

Applications of TVM

- NPV – to be discussed in a subsequent session
- IRR – to be discussed in a subsequent session
- Basic Bond Valuation
- Basic Equity valuation

Principles of bond valuation



- Fundamental principles of valuation
  - Dollar today is higher than dollar received in the future
  - The value of any financial asset equals the PV of its expected cash flows
- Three steps in valuation
  - We must estimate the size/ timing of future cash flows
  - We must determine the appropriate discount rate
  - We must calculate the PV's of the estimated cash flows using the discount rate in step 2

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Basics of Bond Valuation

- The value of the bond is equal to the present value of the cash flows generated by the bond ...

$$P_O = \sum_{t=1}^{T} \frac{C_t}{(1+k)^t} + \frac{Face}{(1+k)^T}$$

Where:

C_t = coupon payment at time t

r = yield to maturity required by the market

Face = face value of the bond to be repaid at the maturity date T

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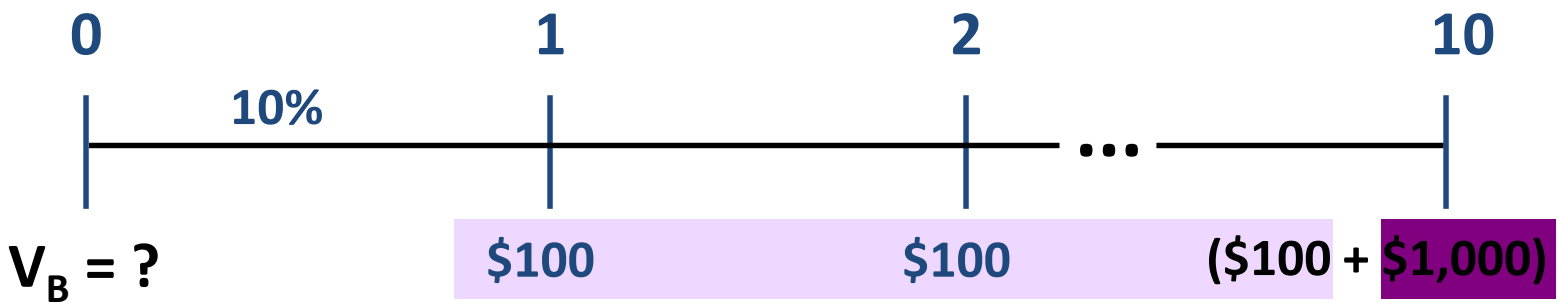
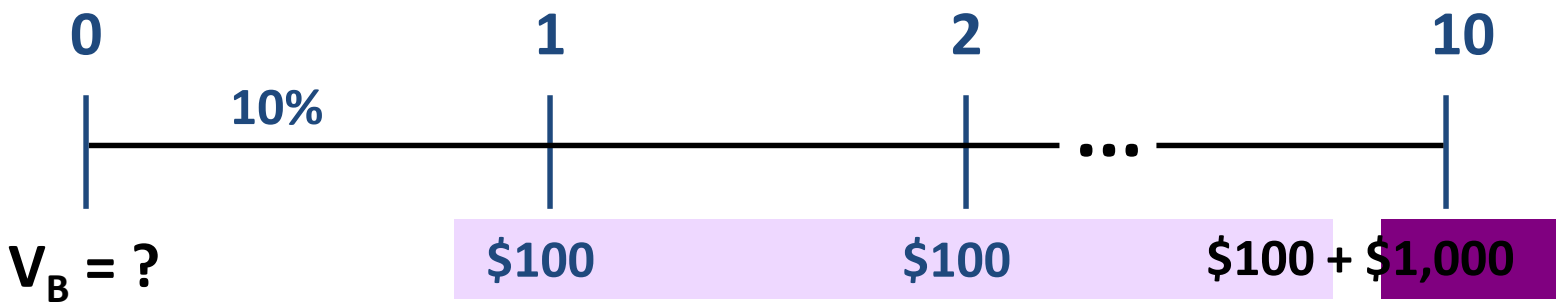
Bond Valuation Example

What's the value of a 10-year, 10% coupon bond if $k_d = 10\%$?



Bond Valuation Example

What's the value of a 10-year, 10% coupon bond if $k_d = 10\%$?



$$V_B = \frac{\$100}{(1+k_d)^1} + \ldots + \frac{\$100}{(1+k_d)^{10}} + \frac{\$1,000}{(1+k_d)^{10}}$$

$$= \$90.91 + \ldots + \$38.55 + \$385.54$$

$$= \$1,000.$$

Valuing preferred Stock

- **Valuation of <u>preferred stock</u>**

  - Relatively easy because dividend is fixed and the preferred life's is infinite (perpetuity)

  V = Dividend/Kp, where *Kp = required return on preferred stock*

- Assume a preferred stock A has a$100 par value and a dividend of $8 a year. Because of inflation, uncertainties and tax advantage, the required rate of return is 9%. Thus, the value of this preferred stock A is……

  V = 8/0.09 = $88.89

  - If the market price of preferred stock A is $95, would you buy it? How about if it's $80?

Valuing common stock using dividend discount models

- **Valuation of <u>common stock</u>**
  - More difficult than bonds and preferred stock because the size and timing of future cash flows are highly uncertain and the required rate of return or discount rate is also unknown
  - The basic model used to value common stock is called the dividend discount model (DDM)
- **DDM determines value of a stock by calculating the PV of the future stream of dividends**

  $V = \sum Dt / (1 + K)$, where *Dt = dividend during period t*

  K = required return
- If an investor sells the stock, the purchaser is buying the remaining dividend streams

DDM for a one-year holding period:

- **Valuation of common stock (<u>one-year holding period</u>)**

  - Value = PV of all any dividends + PV of the selling price

  $V = D_1 / (1 + K) + SP / (1 + K)$,

  Where: SP = selling price; D_1 = dividend to be received at end of year 1

- PVIF = 1/(1+K)^t

- What is the value of a stock that last year paid a $1 dividend. You think next year's dividend will be 10% higher and the stock will be selling for $25 at year-end. The required return or the discount rate is 11%.

  - Solve for D_1 = $1 (1.10) = 1.10

  - Substitute all given variables to the equation = TIME VALUE OF MONEY

  V = 1.10 / (1 + 0.11)^1 + 25 / (1 + 0.11)^1

  V = 0.99 + 22.52 = $23.51 * no. of equity share = intrinsic value of the equity

  - Will you buy the stock if the market price is $25? How about $20?

DDM for a multiple-year holding period:

- **Valuation of common stock (<u>multi-year holding period</u>)**
  - Value = PV of all any dividends + PV of the selling price

$V = D_1 / (1 + K) + D_2 / (1 + K)^2 + SP / (1 + K)^2$ … 2-period

$V = D_1 / (1 + K) + D_2 / (1 + K)^2 + D_3 / (1 + K)^3 + SP / (1 + K)^3$ … 3-period

- Assuming a holding period of three years with the following estimated dividend payments: Year 1 = \$1.10; Year 2 = 1.25 and Year 3 = 1.50. Assume an estimated sale price of \$57, three years from now. Required rate of return is 15%.

$V = 1.10 / (1 + .15) + 1.25 / (1 + .15)^2 + 1.50 / (1 + .5)^3 + 57 / (1 + .15)^3$

V = \$40.37

- Will you buy the stock if the market price is more than \$40.37? How about if it is less than?

DDM for infinite period returns

- **Valuation of common stock (<u>infinite period DDM</u>)**
  - Value = PV of all any dividends + PV of the selling price
  - This is also called the constant growth DDM
  - Assumes an estimation of future dividend payment for an infinite period

    $V = D_1 / (k - g)$ … *g = growth rate*

- What is the value of the stock that paid $2 dividend last year if dividends are expected to grow at 5% forever? The required rate of return is 12%
  - Solve for D_1 = $2 (1.05) = 2.10, then …

    V = 2.10 / (.12 - .05) = $30
  - The relationship between k and g is critical. What happens to the stock if the difference widens? How about if it narrows?

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Exercise Problems

1.5 Time Value of Money : Problems

Problem 1: If you invest Rs. 5000 today at a compound interest of 9 per cent, what will be its future value after 75 years?

Problem 2: If the interest rate is 12 per cent, what are the doubling period as per the rule of 72 and the rule of 69 respectively?

Problem 3: A borrower offers 16 per cent nominal rate of interest with quarterly compounding. What is the effective rate of interest?

1.5 Time Value of Money : Problems

Problem 4: -Fifteen annual payments of Rs. 5000 are made into a deposit account that pays 14 per cent interest per year. What is the future value of this annuity at the end of 15 years.

Problem 5: A finance company advertises that it will pay a lump sum of Rs. 44650 at the end of five years to investors who deposit annually Rs. 6000 for 5 years. What is the interest rate implicit in this offer?

Problem 6: What is the present value of Rs. 1000000 receivable 60 years from now, if the discount rate is 10 percent?

Mini case

Manoj wants to purchase a flat in a new project costing Rs. 95 Lakhs. The builder has an arrangement with a bank, where the bank on behalf of its housing loan borrower, would pay the entire cost of the flat in one lumpsum, once the project reaches the halfway mark, which is expected to be in 3 years. The bank has offered Manoj a housing loan after three years with a down payment of 15 per cent at an effective interest rate of 9 per cent per annum payable in equated monthly instalments (EMI) over a period of 20 years. Manoj has Rs. 14 lakhs which can be kept as a fixed deposit in the bank for 3 years and he can spare an amount not more than Rs. 50000 per month as a recurring deposit in the bank for three years. He understands from the bank that the compound interest rate for the fixed deposit would be 2 percent per quarter and 0.7 per cent per month for the recurring deposit. Before venturing into signing documents with the bank, he wants to be sure that his monthly outgo would be limited to less than Rs. 50000 and has asked your help in ascertaining the following:

Mini Case

1. What would be the EMI on a down payment of 15 per cent of the flat cost?

2.What should be the loan amount (rounded to nearest lakhs) if the EMI should be with Rs.50000?

3.If he goes for a loan amount as in (2), keeping his savings of Rs. 14 lakhs as fixed deposit for 3 years and depositing monthly instalments for 3 years, what would be the monthly recurring deposit instalment?

THANK YOU!!!